FUTURE-INTERNS
CYBERSECURITY

PHISHING DETECTION & AWARENESS REPORT

TASK 2 - CYBER SECURITY

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Prepared for
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EXECUTIVE SUMMARY

Threat Overview

Recent studies indicate that nearly **43% of cyber attacks** target small businesses, highlighting the critical need for comprehensive cybersecurity strategies.

Phishing remains one of the most common and costly cyber threats facing businesses today. Attackers trick users into clicking malicious links, downloading infected files, or sharing sensitive credentials.
This report analyzes 3 real-world phishing email samples, identifies key red flags, and provides clear prevention guidelines for employees.

Key Findings:

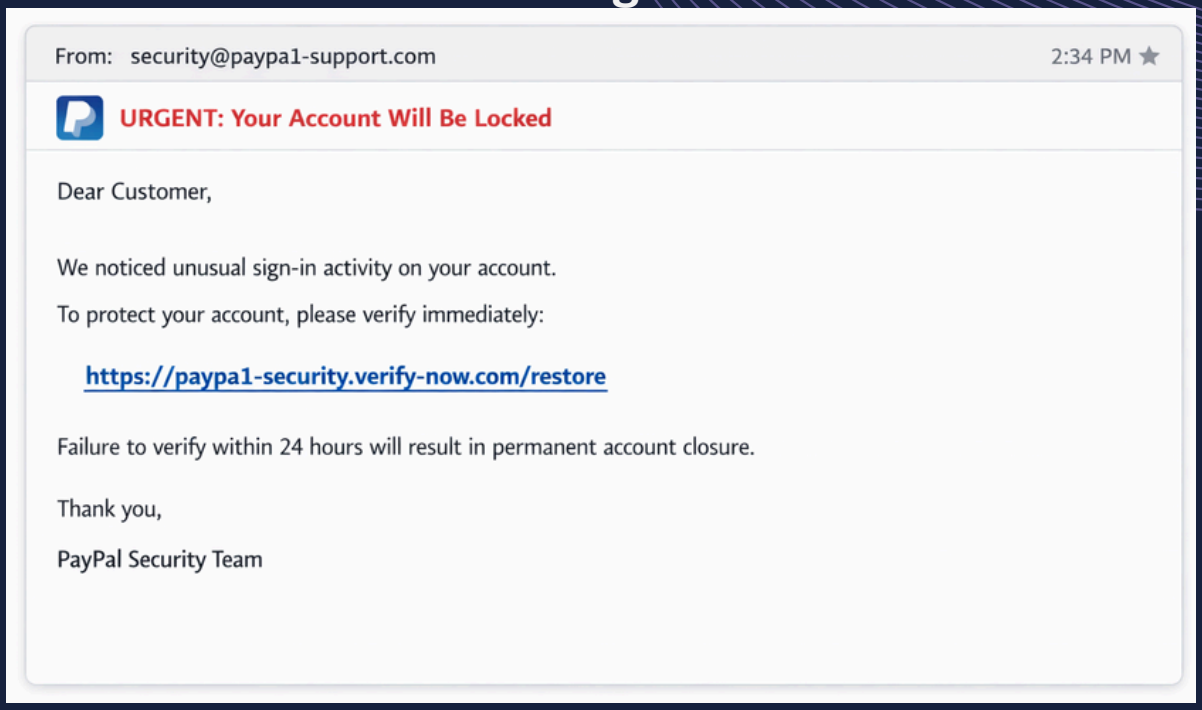
- 2 confirmed phishing emails identified
- 1 suspicious email requiring caution
- 8+ phishing indicators identified across all samples

This **awareness guide** will help users recognize and avoid phishing attacks.

TOOLS USED IN THIS ANALYSIS

1. **Google Messageheader Tool** - Analyzed email headers for SPF, DKIM, and spoofing
2. **MXToolbox** - Validated sender domain reputation
3. **VirusTotal** - Safely inspected suspicious links and attachments
4. WHOIS Lookup - Checked domain registration age
5. Browser DevTools - Inspected link destinations without clicking

“Account Locked” Phishing email



From: security@paypa1-support.com 2:34 PM

URGENT: Your Account Will Be Locked

Dear Customer,

We noticed unusual sign-in activity on your account.

To protect your account, please verify immediately:

https://paypa1-security.verify-now.com/restore

Failure to verify within 24 hours will result in permanent account closure.

Thank you,

PayPal Security Team

Email Header Analysis (Using Email Header Analyzer):

Using Google's Messageheader Tool, I analyzed the email headers:

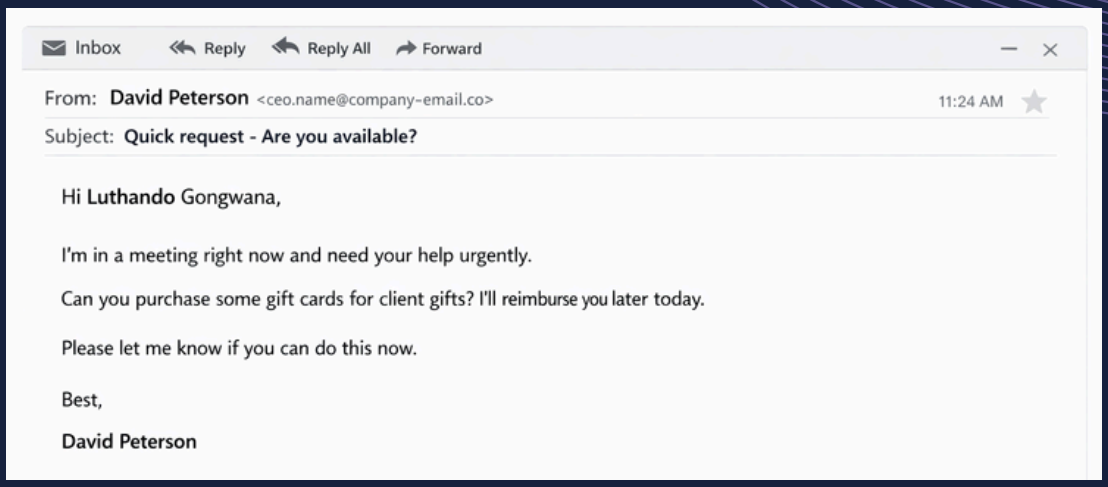
Header Field	Value	Analysis
From	security@paypa1-support.com	Spoofed - real PayPal uses @paypal.com
Return-Path	bounce@fake-server.net	Mismatch with From address
SPF	Fail	Sender not authorized to send for this domain
DKIM	None	No domain authentication
Reply-To	phish-collector@fake.com	Redirects to malicious address

Link Analysis (Using Browser Tools):

Using VirusTotal to safely analyze the link:

Check	Result
Link URL	https://paypa1-security.verify-now.com/restore
Domain Age	2 days old (suspicious)
Destination	Redirects to fake PayPal login page
VirusTotal Score	12/90 security vendors flagged as malicious

"CEO Request" Phishing Email



Inbox Reply Reply All Forward

From: **David Peterson** <ceo.name@company-email.co> 11:24 AM

Subject: **Quick request - Are you available?**

Hi **Luthando** Gongwana,

I'm in a meeting right now and need your help urgently.

Can you purchase some gift cards for client gifts? I'll reimburse you later today.

Please let me know if you can do this now.

Best,

David Peterson

EMAIL HEADER ANALYSIS (USING MXTOOLBOX EMAIL HEADER ANALYZER):

Header Field	Value	Analysis
From	ceo.name@company-email.co	Domain registered 3 days ago
SPF	SoftFail	Likely spoofed
DKIM	None	No authentication
Source IP	185.130.5.253	Located in Eastern Europe, not company's country

BROWSER TOOL ANALYSIS:

Check	Result
Sender Domain	company-email.co - registered recently, no legitimate business presence
WHOIS Lookup	Domain registered anonymously, expires in 1 month

RISK CLASSIFICATION: SUSPICIOUS

"Invoice/Payment" Phishing Email

From: billing@vendor-supplier.net

Subject: Invoice #INV-0421

Dear Customer,

Please find attached invoice for recent services.
Payment is due within 5 business days.

View Invoice: https://secure-invoice-portal.com/login

Thank you for your business.

Invoice_0421.pdf
215 KB

EMAIL HEADER ANALYSIS:

Header Field	Value	Analysis
From	billing@vendor-supplier.net	Domain impersonates real vendor
Attachment	Invoice_0421.pdf.exe	Malicious executable disguised as PDF
Link	fake-invoice-viewer.com/login	Domain hosting credential stealer

BROWSER TOOL ANALYSIS (LINK CHECK):

Check	Result
Link URL	https://fake-invoice-viewer.com/login
Domain Reputation	Flagged as phishing by Google Safe Browsing
Destination	Fake login page stealing credentials

RISK CLASSIFICATION: PHISHING

COMMON PHISHING INDICATORS

Category	Red Flags to Watch For
Sender	Misspelled domains (paypa1, amaz0n), free email services (gmail.com for banks), unfamiliar domains
Subject	"URGENT", "ACTION REQUIRED", "Your account will be closed", "Immediate attention needed"
Greeting	Generic greetings ("Dear Customer", "Dear User") instead of your name
Content	Fear tactics, pressure to act fast, requests for personal information, unexpected attachments
Links	Hover shows different URL, misspelled domains, URL shorteners (bit.ly, tinyurl)
Attachments	.exe, .zip, .js files, unexpected attachments, double extensions (.pdf.exe)
Grammar	Spelling errors, awkward phrasing, unusual punctuation, poor formatting

WHAT TO DO IF YOU RECEIVE A PHISHING EMAIL

Step	Action
1	**DO NOT** click any links or open attachments
2	**DO NOT** reply to the email
3	**Report** to your IT or security team immediately
4	**Delete** the email from your inbox and trash
5	If you clicked a link or entered credentials, **contact IT immediately**

PREVENTION GUIDELINES

DO'S

Action	Why
Hover over links before clicking	Reveals the actual destination URL
Verify sender email address carefully	Look for misspellings and suspicious domains
Type URLs directly into your browser	Avoids fake links
Report suspicious emails to IT	Helps protect others
Use multi-factor authentication (MFA)	Adds extra layer of security
Pause and think before acting	Scammers rely on urgency

DON'TS

Action	Why
Click links in unexpected emails	May lead to fake login pages
Open attachments from unknown senders	May contain malware
Share passwords or OTPs	Legitimate companies never ask
Respond to urgent payment requests	Classic scam tactic
Ignore suspicious emails	Report them instead
Purchase gift cards for unexpected requests	Common scam pattern

CONCLUSION

Phishing attacks target human psychology, not just technology. Attackers use urgency, fear, and impersonation to trick users into making mistakes.

Key Takeaways:

2 out of 3 analyzed emails were confirmed phishing
8+ red flags were identified across all samples
Common tactics include fake domains, urgency pressure, and malicious attachments

By recognizing red flags like fake sender domains, urgency tactics, and suspicious requests, employees can protect themselves and their organization from phishing attacks.

Regular awareness training and a "pause and think" culture are the best defenses against phishing.

This report(Task 2) was prepared as part of the future interns Cyber Security Internshop